

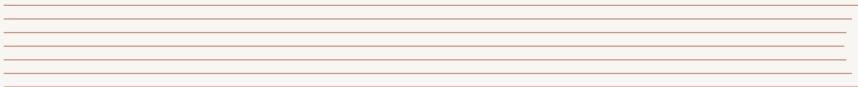
MISSION: POWER UP THE FHR

An Introduction to Reactor Dynamics

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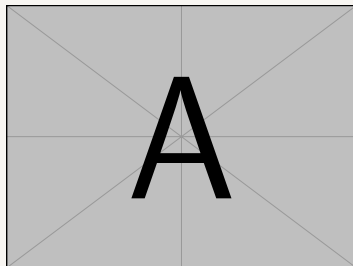
MISSION BRIEFING: GRID DEMAND INCREASE

Currently, our Fluoride-salt-cooled High-temperature Reactor (FHR) is operating stably, supplying baseload power to the grid.

The Situation: Grid demand has just spiked. The grid operator requests an immediate increase in power output.

Your Objective

Increase reactor thermal power from **30 MWth** to **60 MWth** and stabilize it there.



OPERATIONAL CONSTRAINTS & SAFETY LIMITS

Nuclear reactors are powerful systems that require precise control. We cannot just "floor the accelerator."

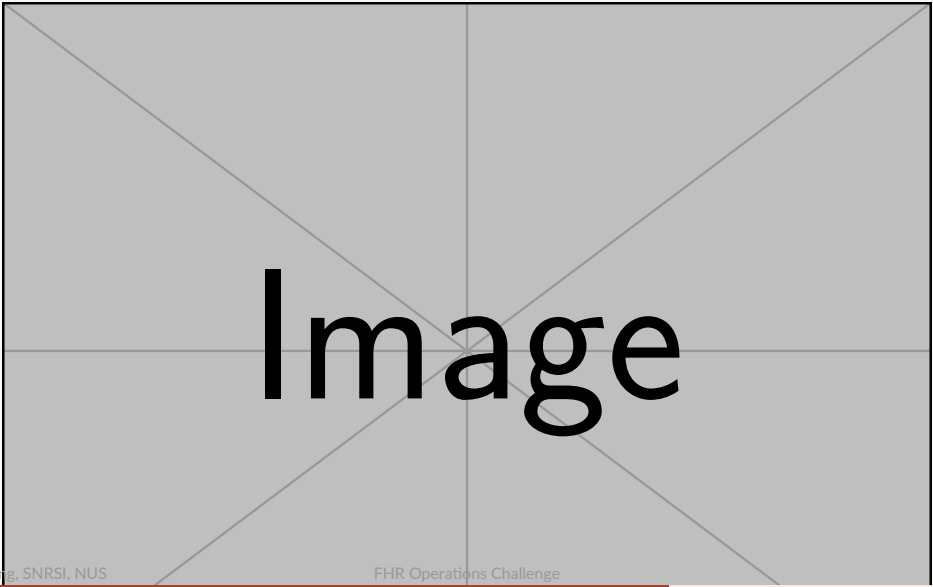
The Rules of Engagement:

- ⦿ You have full control over the reactor's control rods (reactivity).
- ⦿ You must aim for exactly **60 MWth**.

■ SAFETY LIMIT (FAILURE CONDITION)

If reactor power exceeds **1000 MWth** at any point during the maneuver, you have violated the reactor's safety limits and failed the mission.

YOUR INTERFACE: THE FHR CONTROL PANEL



Image

Volunteers Needed!

Who thinks they have the steady hand to bring us to 60 MWth without exceeding 1000 MWth?

For the rest of the class (Observers): Watch closely. Don't just look at the power level.

- ⦿ How does the reactor respond immediately after they move the controls?
- ⦿ What happens to the power when they *stop* moving the controls?
- ⦿ Does the reactor feel "heavy" (slow to move) or "twitchy" (fast to move)?

DEBRIEF: INTUITION VS. REALITY

We just saw a few attempts. It wasn't as easy as turning up a volume knob on a radio.

Discussion Questions:

- ① Did anyone overshoot the 60 MWth target? Why did that happen?
- ② Did you notice the power continuing to change even after you put the controls back to neutral?
- ③ If you added too much reactivity, did it feel like the reactor was "running away" from you?

Why does the reactor have "momentum"? Why doesn't it stop instantly?

To answer this, we need to look at the physics of neutron populations.